

- 8 (a) e.g. infinite (voltage) gain
 infinite input impedance
 zero output impedance
 infinite bandwidth
 infinite slew rate
 (any three, 1 each) B3 [3]
- (b) (i) negative (feedback) B1 [1]
- (ii) 1 gain $(= 5.8/0.069) = 84$ B1 [1]
- (ii) 2 gain $= 1 + 120/X$ C1
 $84 = 1 + 120/X$
 $X = 1.45 \text{ k}\Omega$ A1 [2]
- (iii) gain increases OR bandwidth reduced OR output increases B1 [1]

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9 (a) X ray beam directed through body onto detector (plate)

P1